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Beyond the third lumbar vertebra (L3): Thoracic computed tomography-derived muscle mass and quality assessment as a practical alternative for body composition analysis

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ABSTRACT

Background: Assessing body composition, particularly muscle mass and quality, is crucial in managing advanced non-small-cell lung cancer (NSCLC) patients. While the third lumbar vertebra (L3) computed tomography (CT) scans are the reference standard, thoracic CTs are more commonly performed in clinical practice.
Objective: This study evaluated T12 CT-derived muscle measurements as an alternative to L3, correlating them with dual-energy X-ray absorptiometry (DXA) and handgrip strength (HGS).
Methods: We conducted a cross-sectional study of patients with stage IIIB or IV NSCLC. Skeletal muscle area (SMA), skeletal muscle index (SMI), skeletal muscle radiodensity (SMD), and intermuscular adipose tissue (IMAT) were measured at T12 and L3 using CT. DXA measured total lean soft tissue (LST) mass, trunk LST, and body fat mass. HGS was assessed via a dynamometer. Correlations, linear regression, and Bland–Altman analyses were performed.
Results: We included 332 participants (69.3% men; mean age, 57.3 ± 8.5 y). Strong correlations were found between T12 and L3 SMA (men: r = 0.81; women: r = 0.87) and SMI (men: r = 0.81; women: r = 0.89) (all $P < 0.001$). T12 SMA and SMI exhibited significantly stronger correlations with trunk LST than corresponding L3 measures (SMA: men, r = 0.82 vs. 0.67; women, r = 0.82 vs. 0.72; SMI: men, r = 0.70 vs. 0.53; women, r = 0.73 vs. 0.64; all $P < 0.001$). SMD correlated well between T12 and L3 (men: r = 0.80, $P < 0.001$; women: r = 0.82), but IMAT showed poor correlation (men: r = 0.67, $P < 0.001$; women: r = 0.52) (all $P < 0.001$). Linear regression showed T12 SMA/SMI predicted L3 SMA/SMI, with better R^2 in women (all $P < 0.001$). Bland–Altman analysis showed good agreement for SMI and SMD, but poor agreement for IMAT. Multiple linear regression showed T12 SMA (β = 0.108, R^2 = 0.751) and SMI (β = 0.245, R^2 = 0.610) were stronger predictors of trunk LST than L3 SMA (β = 0.069, R^2 = 0.583) and SMI (β = 0.144, R^2 = 0.459) (all $P < 0.001$). Similarly, T12 SMA (β = 0.238, R^2 = 0.474) and SMI (β = 0.694, R^2 = 0.513) better predicted HGS than L3 parameters (all $P < 0.001$).
Conclusions: T12 CT-derived SMA and SMI are reliable alternatives to L3 for assessing muscle mass in advanced NSCLC, showing strong agreement with L3 and DXA trunk LST. T12 SMD is a suitable substitute for L3 SMD, but T12 IMAT is not. These findings support incorporating T12-level muscle assessment into clinical research as a viable alternative, leveraging the widespread use of thoracic CT scans.

Abbreviations: ALM, appendicular lean mass; AWGS, Asian Working Group for Sarcopenia; BFM, body fat mass; BMI, body mass index; CI, confidence intervals; CT, computed tomography; DXA, dual-energy X-ray absorptiometry; ECOG PS, Eastern Cooperative Oncology Group performance status; HGS, handgrip strength; HU, Hounsfield Unit; IMAT, intermuscular adiposity tissue; L3, the 3rd lumbar vertebra; LST, lean soft tissue; NSCLC, non-small cell lung cancer; SD, standard deviation; SMA, skeletal muscle cross-sectional area; SMD, skeletal muscle radiodensity; SMI, skeletal muscle index (skeletal muscle cross-sectional area/height^2); T12, the 12th thoracic vertebra

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Introduction

Sarcopenia, characterized by the progressive loss of skeletal muscle mass and function, has become a growing global concern with aging populations and rising rates of chronic disease [1,2]. In patients with lung cancer, the prevalence of sarcopenia is notably high, ranging from 43% to 52% depending on the diagnostic criteria, the specific population studied (e.g., stage of disease, treatment received), and the assessment method used [3]. This prevalence is comparable to that observed in other solid tumors, highlighting the importance of addressing sarcopenia across various cancer types [4–6]. Sarcopenia in patients with cancers is not only associated with increased chemotherapy toxicity and postoperative complications but also serves as an independent predictor of overall survival [7–10].

Beyond muscle quantity, composition is increasingly recognized as key to muscle function [1]. Computed tomography (CT) serves as a reference standard for body composition, measuring both muscle area and composition via tissue attenuation (Hounsfield Units, HU) [1,11]. Two key indicators of muscle composition derived from CT are skeletal muscle radiodensity (SMD), which reflects the X-ray attenuation properties of muscle tissue, and intermuscular adipose tissue (IMAT), which represents fat infiltration within the muscle [11]. Low SMD and high IMAT indicate poor muscle composition and have been linked to adverse outcomes, often surpassing muscle mass in prognostic value [12–14]. In lung cancer, low SMD is associated with greater chemotherapy toxicity and reduced survival [12,15].

Currently, CT-derived measurements at the third lumbar vertebra (L3) are the reference standard for assessing muscle mass and composition in clinical research [16–18]. Muscle mass is quantified by skeletal muscle area (SMA) and skeletal muscle index (SMI), while composition is assessed via IMAT and SMD [16,17]. However, chest CT scans in lung cancer patients often omit the L3 level, leading to exclusion of up to one-third of patients in some studies [19,20]. To address this, alternative sites like the twelfth thoracic vertebra (T12) are being investigated. T12 measurements correlate well with those at L3 [21–23] and offer practical advantages: they are captured in both chest and abdominal CTs, require no additional imaging, and reduce radiation exposure—making T12 a viable alternative.

While T12 and L3 muscle measurements correlate, it's unclear which site better reflects overall body composition, especially trunk muscle mass [24–26]. The association between these measurements and handgrip strength (HGS)—a key marker of muscle function and sarcopenia diagnosis [1]—also requires further study. Specifically, direct comparisons of the predictive value of T12 and L3 parameters for HGS are lacking. Data on muscle composition indicators such as SMD and IMAT at T12 are limited, and whether they differ from those at L3 is unknown. Given that T12 primarily involves the erector spinae muscles while L3 includes the psoas and others, these anatomical differences might affect muscle composition assessment [13]. This study aims to address these gaps by comparing T12 and L3 muscle parameters, including muscle mass and composition, and their relationships with dual-energy X-ray absorptiometry (DXA)-derived body composition and HGS in patients with advanced non-small cell lung cancer (NSCLC).

Methods

Study population

This single-center, retrospective, cross-sectional study was approved by the Biomedical Research Ethics Committee of West China Hospital, Sichuan University (Approval No.: 2016-331), and all participants provided written informed consent. Between August 2017 and May 2019, patients with NSCLC were consecutively recruited from the Oncology Department of Shangjin Nanfu Hospital, Sichuan University [27]. Inclusion criteria were: 1) pathologically confirmed stage IIIB or IV NSCLC; 2) hospitalized for the initiation of first-line chemotherapy for the first time; and 3) with both thoracic and abdominal CT images. All patients were admitted for baseline evaluation prior to the initiation of first-line chemotherapy. Exclusion criteria were: 1) pacemaker implantation; 2) history of other malignancies; 3) poor CT image quality, anatomical distortions (e.g., chest wall edema), or missing muscle areas on CT images; 4) visible edema; or 5) Eastern Cooperative Oncology Group (ECOG) performance status score greater than 2.

Clinical and anthropometric data collection

Within 48 h of admission, trained nurses recorded the following clinical data: age, sex, smoking status (never or ever), histological type (adenocarcinoma, squamous cell carcinoma, or large cell carcinoma), cancer stage (IIIB or IV), ECOG performance status score, number of chemotherapy cycles, and chemotherapy regimen.

Serum creatinine, serum albumin, and hemoglobin levels were measured within 24 h of admission. The updated Charlson Comorbidity Index was used to assess the number and severity of significant comorbidities [28]. Body height and weight were measured using standard methods, and body mass index was calculated.

CT scanning and image analysis

Chest and abdominal CT scans were performed within 48 h of admission using a 16-slice spiral CT scanner (Brilliance; Philips Healthcare) with a slice thickness of 5 mm. Importantly, non-contrast CT images were used for all analyses, as previous studies have reported that contrast agents can significantly influence SMD measurements [16]. All patients were admitted for baseline evaluation prior to the initiation of first-line chemotherapy.

Segmentation was performed semiautomatically using Mimics version 21.0 (Materialise, Leuven, Belgium) to analyze cross-sectional CT images at the T12 and L3 levels. The axial slice located at the midpoint of the vertebral body was selected for analysis, defined as the image showing the maximal cross-sectional area of the vertebral body with clearly visible bilateral pedicles. On a single CT image, all visible skeletal muscles were segmented (threshold −29 to +150 HU) [16]. At T12, this included the erector spinae, latissimus dorsi, rectus abdominis, external and internal obliques, and intercostal muscles; at L3, it included the psoas, erector spinae, rectus abdominis, external and internal obliques, and quadratus lumborum. SMA was measured automatically after the segmentations, and SMI was calculated as SMA/height^2. SMD was measured as the average CT value of all voxels within the segmented area. IMAT was defined as tissue within the −190 to −30 HU range (Supplementary Fig. 1) [16]. All measurements were performed by the same trained radiologist, who was blinded to all clinical data, using uniform segmentation parameters and protocols for both vertebral levels.

DXA for body composition

Body composition was assessed using DXA on the same day as the CT scans. Participants underwent whole-body scans using a DXA scanner (Discovery DXA system, Hologic Inc., Bedford, MA, USA) while positioned supine. The APEX software (version 4.0) was used to obtain the following parameters: body fat mass, total lean soft tissue (LST), and trunk LST. To ensure measurement accuracy and consistency, all scans were performed by trained technicians following standardized operating procedures. Daily instrument calibration was performed, and participants were scanned in light clothing without any metallic objects.

HGS measurement

HGS was measured on the same day as the CT scan using a digital handgrip dynamometer (EH101; Xiangshan Inc., Guangdong, China). Participants were seated with their upper arm naturally adducted, elbow flexed at 90 degrees, and forearm and wrist in a neutral position. At the tester's command, participants squeezed the dynamometer with maximum force and maintained it for 2 to 3 s. Each hand was measured alternately three times, with a rest of at least 30 s between measurements. The maximum value for each hand was recorded. The highest value from both hands was used as the final HGS value, accurate to 0.1 kg. All measurements were performed by trained nurses.

Statistical analysis

Statistical analysis was performed using R software version 4.0.3 and SPSS version 26.0. The Shapiro-Wilk test was used to assess the normality of continuous variables. Data are presented as mean ± standard deviation for normally distributed continuous variables and as counts (percentages) for categorical variables.

Pearson correlation analysis was used to evaluate the correlation between CT-derived measurements, DXA parameters, and HGS. In addition, correlation heatmaps stratified by sex were generated. The strength of correlation coefficients was interpreted as follows: $|r| \geq 0.8$ as strong, $0.5 \leq |r| < 0.8$ as moderate, $0.3 \leq |r| < 0.5$ as weak, and $|r| < 0.3$ as very weak or no correlation.

Scatter plots and linear regression analysis were employed to assess the following relationships: 1) between T12 and L3 measurements; 2) between T12 and L3 measurements and DXA-determined total LST and trunk LST; and 3) between T12 and L3 measurements and HGS. All analyses were performed separately by sex, and regression equations and coefficients of determination (R^2) were calculated. Bland−Altman analysis was used to assess the agreement between T12-predicted and L3-measured values, calculating bias (mean difference) and 95% confidence interval (95% CI).

Univariable and multiple linear regression analyses were used to examine the relationship between CT-derived parameters and trunk LST and HGS. In the multivariable models, Model 1 was adjusted for age (continuous) and sex, while Model 2 was further adjusted for histological type (adenocarcinoma or squamous cell carcinoma) and cancer stage (IIIB or IV). Regression results are presented as regression coefficients (β), 95% 95% CI, P values, and coefficients of determination (R^2).

To assess the agreement between T12- and L3-derived measurements, we applied multiple complementary statistical methods. Intraclass correlation coefficients (ICCs) were calculated using a two-way mixed-effects model with absolute agreement to quantify reliability across vertebral levels. Standard errors of the estimate were derived from the regression models to reflect the average prediction error.

All statistical tests were two-sided, and a P value <0.05 was considered statistically significant.

Results

Patient characteristics

A total of 332 patients with advanced NSCLC were included in this study, comprising 230 men (69.3%) and 102 women (30.7%), with a mean age of 57.3 ± 8.5 y. The cohort consisted primarily of adenocarcinoma (187 patients, 56.3%) and squamous cell carcinoma (145 patients, 43.7%). Notably, the proportion of adenocarcinoma was significantly higher in women (76.5%, 78/102) than in men (47.4%, 109/230; $P < 0.001$). The majority of patients presented with stage IV disease (63.3%, 210/332) and an ECOG performance status of 0 to 1 (80.4%, 267/332) (Table 1).

Agreement between T12 and L3 muscle measurements

Strong correlations were observed between T12 and L3 measurements for both SMA and SMI in men (r = 0.81 and 0.81, respectively, both $P < 0.001$) and women (r = 0.87 and 0.89, respectively, both $P < 0.001$) (Fig. 1). Linear regression showed that T12 measurements effectively predicted L3 values. Notably, the coefficients of determination were higher in women than in men for both SMA (R^2 = 0.76 vs. 0.66, both $P < 0.001$) and SMI (R^2 = 0.79 vs. 0.66, both $P < 0.001$) (Fig. 2).

Similarly, SMD at T12 and L3 showed strong correlations in both men (r = 0.80, $P < 0.001$) and women (r = 0.82, $P < 0.001$), with similar predictive ability in regression models (R^2 = 0.65 and 0.67, respectively, both $P < 0.001$) (Fig. 1). However, IMAT showed only moderate correlations between the two levels (men: r = 0.67, $P < 0.001$; women: r = 0.52, $P < 0.001$), and T12 IMAT demonstrated limited predictive capacity for L3 IMAT values, especially in women (men: R^2 = 0.45, $P < 0.001$; women: R^2 = 0.27, $P < 0.001$) (Fig. 2).

Bland−Altman analysis revealed a mean bias of −0.15 cm^2/m^2 (95% limits of agreement: −6.8 to 6.5 cm^2/m^2) for SMI and −0.08 HU (95% limits of agreement: −6.2 to 6.1 HU) for SMD between T12 and L3 measurements. However, IMAT exhibited poor agreement, with a mean bias of 0.15 cm^2 (95% limits of agreement: −7.2 to 7.5 cm^2), and particularly larger deviations were observed in the higher value range (>6 cm^2) (Fig. 3).

The standard errors of the estimate for T12-predicted L3 values were low across all parameters, indicating strong predictive performance (Table 2). We further assessed the agreement between T12 and L3 measurements using ICCs. As shown in Table S1, ICCs for SMA, SMI, and SMD all above 0.83, indicating excellent agreement.

Correlation between CT and DXA measurements

T12 SMA demonstrated a strong correlation with DXA-measured trunk LST in both men (r = 0.82, $P < 0.001$) and women (r = 0.82, $P < 0.001$), exceeding the correlation observed between L3 SMA and trunk LST (men: r = 0.67, $P < 0.001$; women: r = 0.72, $P < 0.001$). However, the correlation between T12 SMA and total LST was weaker (men: r = 0.54, $P < 0.001$; women: r = 0.43, $P < 0.001$) compared to its correlation with trunk LST (Fig. 1).

Similarly, T12 SMI exhibited a moderate correlation with trunk LST in both men (r = 0.70, $P < 0.001$) and women (r = 0.73, $P < 0.001$), which was stronger than the correlation between L3 SMI and trunk LST (men: r = 0.53, $P < 0.001$; women: r = 0.64, $P < 0.001$) (Fig. 1).

T12 SMA also showed a moderate correlation with total LST in men (r = 0.54, $P < 0.001$) but only a weak correlation in women

Table 1
Baseline characteristics of the study population

Characteristic	Total (n = 332)	Men (n = 230)	Women (n = 102)	P value
Age, y, mean (SD)	57.3 (8.5)	57.8 (8.0)	56.2 (9.4)	0.118
Ever smoker, n (%)	193 (58.1)	186 (80.9)	7 (6.9)	<0.001
Histologic type, n (%)				
Adenocarcinoma	187 (56.3)	109 (47.4)	78 (76.5)	<0.001
Squamous carcinoma	145 (43.7)	121 (52.6)	24 (23.5)	
Cancer stage, n (%)				
Stage IIIB	122 (36.7)	91 (39.6)	31 (30.4)	0.110
Stage IV	210 (63.3)	139 (60.4)	71 (69.6)	
ECOG PS score, n (%)				
0–1	267 (80.4)	185 (80.4)	82 (80.4)	0.993
2	65 (19.6)	45 (19.6)	20 (19.6)	
Height, cm, mean (SD)	163.3 (7.5)	166.6 (5.4)	156.0 (6.4)	<0.001
Weight, kg, mean (SD)	61.2 (9.7)	63.5 (9.5)	56.0 (8.0)	<0.001
BMI, kg/m^2, mean (SD)	22.9 (2.9)	22.8 (3.0)	23.0 (2.9)	0.589
BMI groups, n (%)				
BMI < 20 kg/m^2	49 (14.8)	36 (15.7)	13 (12.7)	0.483
BMI 20–24.9 kg/m^2	202 (60.8)	135 (58.7)	67 (65.7)	
BMI ≥ 25 kg/m^2	81 (24.4)	59 (25.7)	22 (21.6)	
Body composition variables				
T12 SMA, cm^2, mean (SD)	88.2 (17.7)	96.3 (14.1)	70.0 (9.6)	<0.001
T12 SMI, cm^2/m^2, mean (SD)	32.9 (5.5)	34.7 (5.0)	28.8 (4.2)	<0.001
T12 SMD, HU, mean (SD)	34.9 (5.8)	36.5 (5.0)	31.2 (5.9)	<0.001
T12 IMAT, cm^2, mean (SD)	4.7 (3.3)	4.6 (3.3)	5.1 (3.0)	0.157
L3 SMA, cm^2, mean (SD)	122.9 (23.0)	133.4 (18.4)	99.2 (12.3)	<0.001
L3 SMI, cm^2/m^2, mean (SD)	45.9 (7.1)	48.1 (6.7)	40.8 (6.7)	<0.001
L3 SMD, HU, mean (SD)	34.9 (6.0)	36.3 (5.2)	31.5 (6.1)	<0.001
L3 IMAT, cm^2, mean (SD)	7.7 (4.3)	7.9 (4.5)	7.1 (3.6)	0.098
total LST, kg, mean (SD)	26.6 (5.2)	29.2 (4.6)	22.8 (4.4)	<0.001
trunk LST, kg, mean (SD)	7.5 (1.9)	8.1 (1.9)	6.1 (1.0)	<0.001
Total BFM, kg, mean (SD)	16.2 (5.1)	16.0 (5.1)	16.6 (5.1)	0.315
HGS, kg, mean (SD)	26.6 (6.6)	28.8 (5.5)	21.6 (6.3)	<0.001

ALM, appendicular lean mass; BFM, body fat mass; BMI, body mass index; ECOG PS, Eastern Cooperative Oncology Group performance status; HGS, handgrip strength; HU, Hounsfield Unit; IMAT, intermuscular adiposity tissue; L3, the 3rd lumbar vertebra; LST, lean soft tissue; SD, standard deviation; SMA, skeletal muscle cross-sectional area; SMD, skeletal muscle radiodensity; SMI, skeletal muscle index (skeletal muscle cross-sectional area/height^2); T12, the 12th thoracic vertebra.

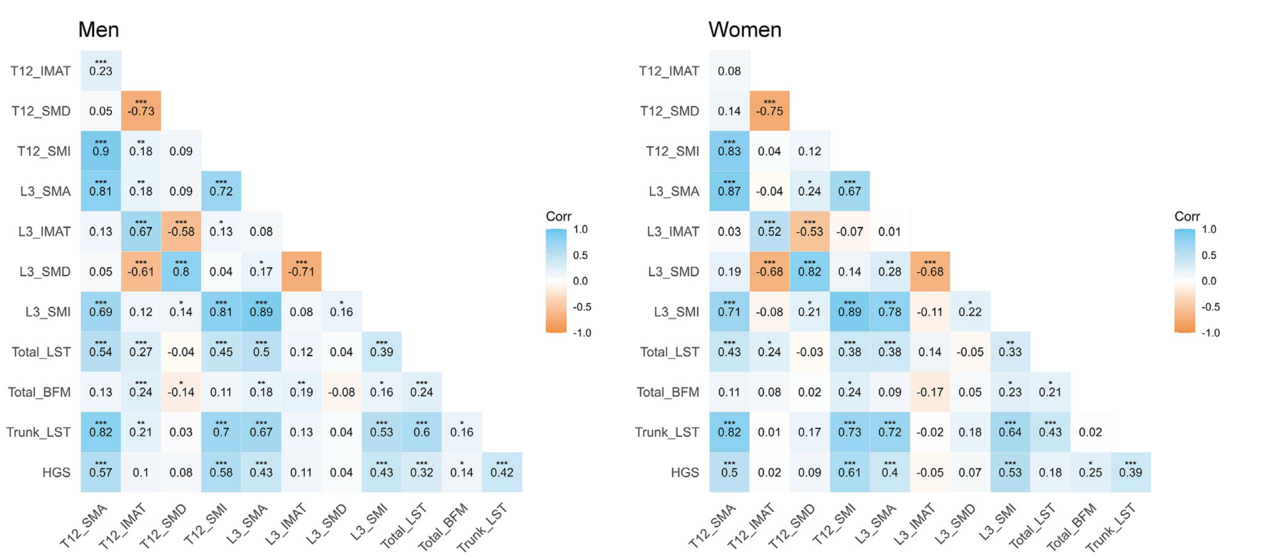

Fig. 1. Correlation heatmaps of CT-derived muscle measurements, DXA parameters, and handgrip strength, stratified by sex. Statistical significance: * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$. Corr, correlation; CT, computed tomography; DXA, dual-energy X-ray absorptiometry; HGS, handgrip strength; IMAT, intermuscular adipose tissue; L3, third lumbar vertebra; SMA, skeletal muscle area; SMD, skeletal muscle radiodensity; SMI, skeletal muscle index; T12, twelfth thoracic vertebra; Total BFM, total body fat mass; total LST, total lean soft tissue; trunk LST, trunk lean soft tissue.

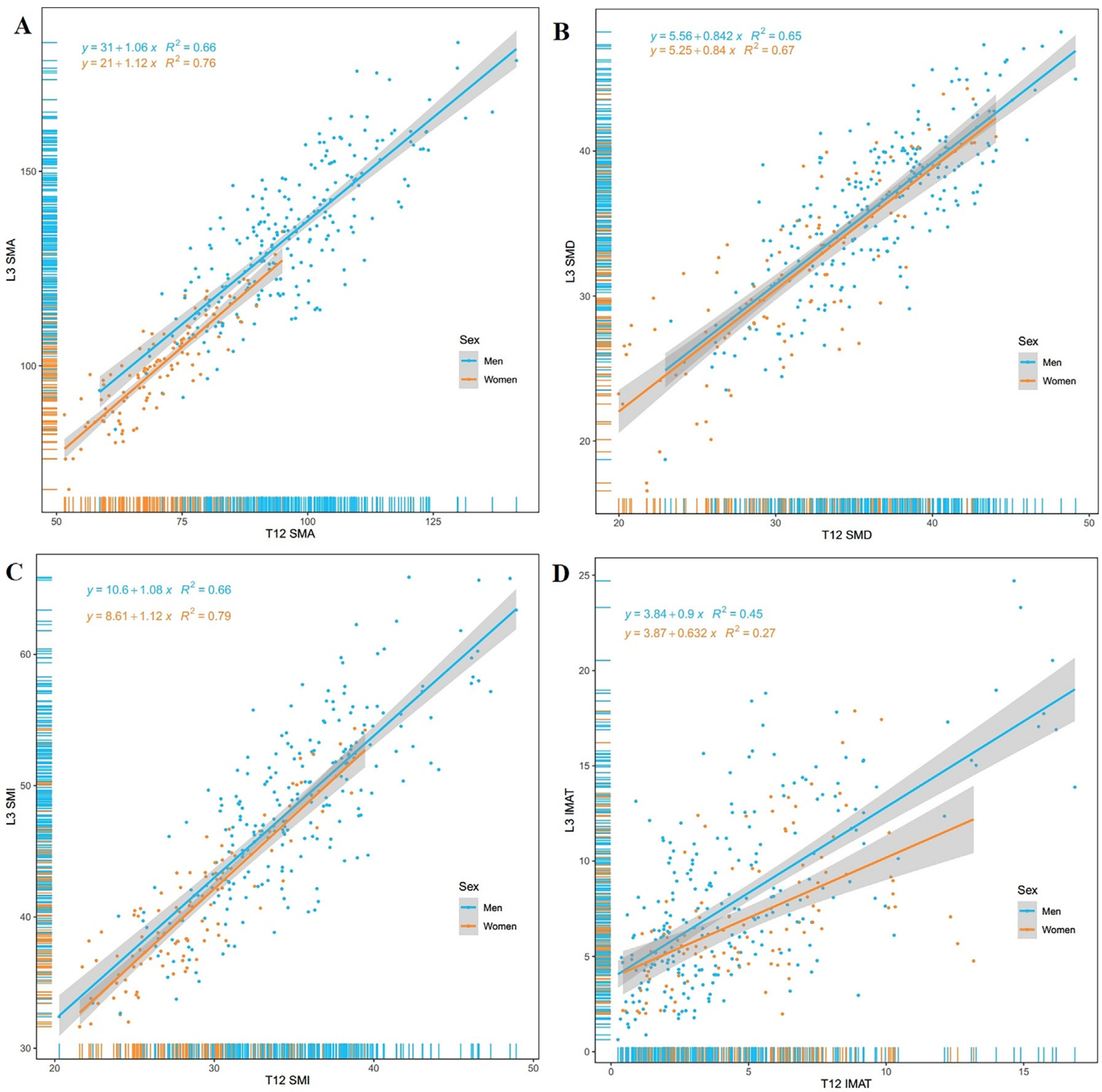

Fig. 2. Scatter plots and linear regression of muscle measurements, SMA (A), SMD (B), SMI (C), and IMAT (D), between T12 and L3 Levels, stratified by sex. IMAT, intermuscular adipose tissue; L3, third lumbar vertebra; R^2, *R*-squared (coefficient of determination); SMA, skeletal muscle area; SMD, skeletal muscle radiodensity; SMI, skeletal muscle index; T12, twelfth thoracic vertebra.

(r = 0.43, P < 0001), both of which were stronger than the correlation between L3 SMA and total LST (men: r = 0.50, P < 0.001; women: r = 0.38, P < 0.001). A similar pattern was observed for the relationship between SMI and total LST in both men and women (Fig. 1).

Correlation between CT measurements and HGS

HGS moderately correlated with T12 SMA in both men (r = 0.57, P < 0.001) and women (r = 0.50, P < 0.001). A similar correlation was observed between HGS and T12 SMI (men: r = 0.58, P < 0.001; women: r = 0.61, P < 0.001) (Fig. 1).

However, the correlations between HGS and L3 SMA were weak (men: r = 0.43, P < 0.001; women: r = 0.40, P < 0.001), as were those with L3 SMI (men: r = 0.43, P < 0.001; women: r = 0.53, P < 0.001) (Fig. 1).

Association between CT measurements, HGS, and trunk LMB

Multiple linear regression analysis, adjusted for age, sex, histological type, and stage, showed that T12 SMA was a stronger predictor of trunk LST (β = 0.108, 95% CI: 0.100–0.116, P < 0.001, R^2 = 0.751) compared to L3 SMA (β = 0.069, 95% CI: 0.061–0.077, P < 0.001, R^2 = 0.583). Similarly, T12 SMI exhibited a stronger association with trunk LST (β = 0.245, 95% CI: 0.218–0.273, P < 0.001, R^2 = 0.610), compared to L3 SMI (β = 0.144, 95% CI: 0.119–0.169, P < 0.001, R^2 = 0.459) (Table 2).

After adjusting for confounders, T12 SMA also demonstrated better predictive power for HGS (β = 0.238, 95% CI: 0.197–0.279, P < 0.001, R^2 = 0.474) than L3 SMA (β = 0.139, 95% CI: 0.104–0.174, P < 0.001, R^2 = 0.383). A similar trend was observed for T12 SMI and L3 SMI in relation to HGS (Table 2).

SMD at either T12 or L3 showed no significant association with trunk LST or HGS after adjusting for confounders (Table 2).

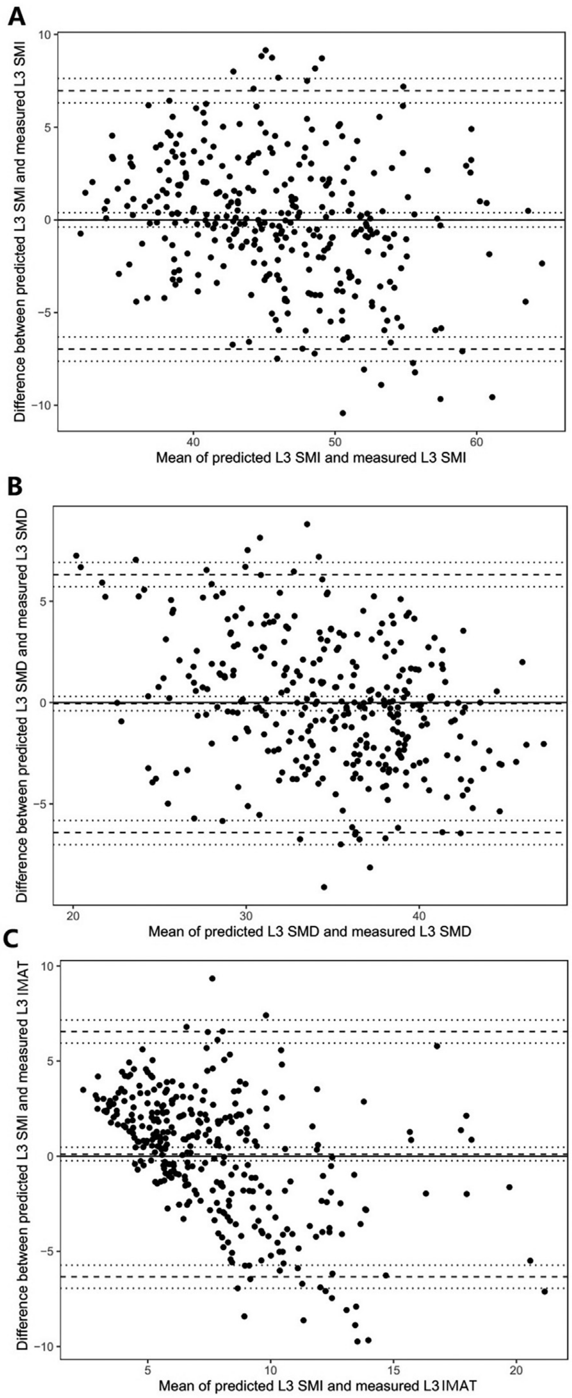

Fig. 3. Bland–Altman plots showing the agreement between T12 and L3 muscle measurements, including SMI (A), SMD (B), and IMAT (C). The thick dashed lines represent the 95% limits of agreement; the surrounding thinner dotted lines indicate their 95% confidence intervals. "Predicted L3" refers to values estimated using linear regression based on T12 muscle parameters. IMAT, intermuscular adipose tissue; L3, third lumbar vertebra; SMD, skeletal muscle radiodensity; SMI, skeletal muscle index; T12, twelfth thoracic vertebra.

Discussion

This study systematically evaluated the utility of T12-level CT measurements in assessing muscle mass and composition in patients with advanced lung cancer. We found that T12-level SMA and SMI strongly correlate with L3-level measurements, and T12 measurements, particularly SMA, show a stronger correlation with DXA-determined trunk LST than L3 measurements. Regarding muscle composition, SMD demonstrated good agreement between the two anatomical locations, whereas IMAT showed a weaker and less reliable correlation. These findings provide important evidence for using T12-level CT measurements, especially SMA and SMI, to assess body composition in lung cancer patients, offering a practical alternative when L3-level scans are unavailable.

CT measurements have become an important tool for assessing body composition in cancer patients, with L3-level measurements widely recognized as the reference standard for evaluating skeletal muscle mass. This recognition is largely due to the findings that a single L3-level CT slice strongly correlates highly with whole-body skeletal muscle volume [18]. However, routine chest CT scans in lung cancer patients often do not include the L3 level, limiting its clinical utility. Our study identified a significant correlation between T12 and L3 SMA and SMI (r = 0.81–0.89), aligning with previous research. For example, Nemec et al. [21] reported a correlation of 0.724 between T12 and L3 SMA in 157 patients undergoing transcatheter aortic valve replacement. Similarly, Kaplan et al. [29] reported a correlation of 0.62 between the paraspinal muscles between T12 and L3 in 316 older trauma patients.

Research comparing different anatomical locations suggests the importance of proximity to the L3 level when selecting alternative measurement locations. For example, Heusden et al. [30] reported a lower but statistically significant correlation between T4 and L3 measurements (r = 0.791), which was weaker than the correlation observed at T12 in our lung cancer cohort. Similarly, studies by Derstine et al. [22] in healthy adults demonstrated that correlation strength decreases as the distance from L3 increases. These findings collectively underscore the suitability of T12 as a viable alternative to L3.

Importantly, our study revealed that T12 SMA correlated more strongly with DXA-determined trunk LST (r = 0.82) than L3 SMA (r = 0.67–0.72). This finding has significant clinical implications, suggesting that T12 may be a better indicator of trunk muscle status, which is particularly relevant in patients with lung cancer. The superior correlation of T12 with trunk LST may be attributed to its anatomical proximity to major trunk muscle groups, more representative muscle composition and distribution at the thoracolumbar junction, and clearer muscle outlines, which potentially reduce measurement error. Our data also show that this correlation is stable across different sexes, further supporting the reliability of T12 measurements.

In terms of muscle composition assessment, this study is the first to systematically compare SMD and IMAT at different anatomical locations. SMD showed good agreement between the two locations (r = 0.80–0.82), a finding with significant implications. Previous studies have shown that low L3 SMD is closely related to poor prognosis [19], poor physical function [31], and increased chemotherapy-induced toxicity [32] in patients with cancer. Our study suggests that T12 SMD measurements can be a reliable alternative, expanding the applicability of SMD assessment in clinical practice, especially when L3 scans are unavailable. However, IMAT showed a much weaker correlation, potentially reflecting the site-specific nature of fat deposition patterns. Furthermore, the predictive deviations of T12 IMAT were larger in the high-value range (>6 cm^2), highlighting the unreliability of T12 IMAT as a substitute for L3 IMAT, particularly in patients with severe muscle fat infiltration.

While the revised European consensus on sarcopenia [1] acknowledges a moderate correlation between muscle mass and strength, our study found this relationship to be more pronounced when assessing muscle mass at the T12 level rather than the L3

Table 2
Association of CT-derived body composition components with trunk lean soft tissue and handgrip strength based on simple and multiple linear regression

	Simple linear regression					Multiple linear regression Model 1					Multiple linear regression Model 2				
	β	95% CI	P	R^2	SEE	β	95% CI	P	R^2	SEE	β	95% CI	P	R^2	SEE
Trunk LST															
T12 SMA	0.093	0.087–0.099	<0.001	0.723	1.022	0.108	0.099–0.116	<0.001	0.746	0.980	0.108	0.100–0.116	<0.001	0.751	0.974
T12 SMI	0.270	0.245–0.294	<0.001	0.588	1.246	0.245	0.217–0.273	<0.001	0.607	1.220	0.245	0.218–0.273	<0.001	0.610	1.220
T12 SMD	0.082	0.048–0.117	<0.001	0.062	1.880	−0.002	−0.041 to 0.036	0.909	0.245	1.691	−0.004	−0.042 to 0.035	0.860	0.246	1.695
L3 SMA	0.064	0.058–0.070	<0.001	0.572	1.269	0.068	0.060–0.076	<0.001	0.576	1.268	0.069	0.061–0.077	<0.001	0.584	1.259
L3 SMI	0.174	0.152–0.197	<0.001	0.412	1.488	0.143	0.118–0.168	<0.001	0.455	1.437	0.144	0.119–0.169	<0.001	0.459	1.436
L3 SMD	0.077	0.042–0.111	<0.001	0.056	1.886	0.002	−0.035 to 0.038	0.087	0.245	1.691	0.002	−0.035 to 0.039	0.908	0.246	1.695
Handgrip strength															
T12 SMA	0.255	0.226–0.285	<0.001	0.463	4.876	0.236	0.195–0.278	<0.001	0.469	4.856	0.238	0.197–0.279	<0.001	0.474	4.856
T12 SMI	0.827	0.731–0.922	<0.001	0.469	4.849	0.692	0.586–0.798	<0.001	0.509	4.677	0.694	0.588–0.800	<0.001	0.513	4.672
T12 SMD	0.315	0.196–0.433	<0.001	0.077	6.396	0.032	−0.098 to 0.162	0.629	0.264	5.728	0.029	−0.103 to 01.60	0.666	0.266	5.737
L3 SMA	0.173	0.148–0.198	<0.001	0.361	5.322	0.137	0.102–0.171	<0.001	0.377	5.268	0.139	0.104–0.174	<0.001	0.383	5.260
L3 SMI	0.540	0.458–0.622	<0.001	0.337	5.420	0.403	0.314–0.493	<0.001	0.406	5.145	0.407	0.317–0.496	<0.001	0.410	5.141
L3 SMD	0.255	0.138–0.373	<0.001	0.052	6.479	0.848	−0.137 to 0.113	0.848	0.263	5.729	−0.008	−0.134 to 0.117	0.896	0.266	5.736

AWGS, Asian Working Group for Sarcopenia; BMI, body mass index; CT, computed tomography; ECOG PS, Eastern Cooperative Oncology Group performance status; HU, Hounsfield Unit; LST, lean soft tissue; SD, standard deviation; SEE, standard error of the estimate; SMA, skeletal muscle cross-sectional area; SMD, skeletal muscle radiodensity; SMI, skeletal muscle index (skeletal muscle cross-sectional area/height^2); T12, the 12th thoracic vertebra.
Model 1: adjustment for age (continuous data) and sex. Model 2: adjustment for age (continuous data), sex, histology type (adenocarcinoma or squamous carcinoma), and cancer stage (IIIB or IV).

level, using HGS as a measure of strength. We observed a stronger correlation between T12 SMI and HGS (r = 0.58−0.61) compared to L3-SMI (r = 0.43−0.53). Although HGS predominantly reflects forearm muscle strength, it is often used as a surrogate for overall muscle strength and functional capacity due to its ease of measurement and strong associations with clinical outcomes [33]. The stronger correlation observed at T12 might be related to the indirect role of trunk muscles, such as the erector spinae and latissimus dorsi (captured at T12), in providing core stability and supporting limb movements during the HGS test. This is supported by studies demonstrating the contribution of trunk muscles to overall physical performance and functional tasks, even those involving the upper limbs [34].

The accuracy of CT image analysis for muscle assessment is affected by several factors, demanding careful consideration and standardized protocols. First, the administration of intravenous contrast agents is an important factor. Contrast enhancement increases muscle CT values, particularly in the arterial phase, and the degree of enhancement is dependent on contrast agent concentration, injection rate, and scan timing relative to injection. Therefore, it is crucial to use either non-enhanced scans or images acquired in the same phase of enhancement (e.g., portal venous phase) for consistent muscle measurements [16]. Second, respiratory motion can affect muscle morphology, especially at the T12 level, where diaphragmatic movement can alter the shape and position of surrounding muscles. Techniques to minimize respiratory artifacts, such as breath-hold instructions or respiratory gating, should be employed when possible [35]. Finally, other factors like slice thickness, reconstruction algorithms, and even the specific CT scanner model can influence the measurements, further emphasizing the need for standardized protocols.

This study has several limitations. First, the cross-sectional design limits causal inferences. While we observed good correlations between T12 and L3 measurements and DXA results, we cannot determine whether this correlation remains stable during disease progression. This limitation is particularly important given that body composition in cancer patients can change dynamically with disease progression and treatment. Second, as a single-center study, our results may be influenced by specific population and measurement conditions. Differences in CT equipment, scanning parameters, and image reconstruction algorithms between centers can affect measurement comparability. Additionally, the patient population at our center may differ from other regions, potentially affecting the generalizability of the results. Third, the study was limited to advanced lung cancer patients, who may have unique body composition characteristics. For example, compared to other cancers, lung cancer patients may have more significant respiratory muscle changes and a more pronounced systemic inflammatory response, which could affect muscle morphology. Therefore, whether the results are applicable to early-stage lung cancer patients or other malignancies requires further investigation. Finally, skeletal muscle and IMAT segmentation were performed using Mimics software with standardized HU thresholds by an experienced radiologist; however, minor inclusion of adjacent non-muscle soft tissues, especially in regions of muscle thinning or low contrast, may have introduced slight measurement bias.

To translate these findings into clinical practice, future research should focus on several key areas. Firstly, large-scale, multi-center prospective studies are essential to validate the clinical utility of T12 measurements in predicting patient outcomes and guiding treatment decisions. This includes investigating their role in early warning systems and treatment monitoring. Secondly, developing and validating user-friendly, automated measurement tools based on artificial intelligence and deep learning will be crucial for seamless integration into routine clinical workflows. These tools could enhance efficiency and potentially improve the sensitivity of muscle assessments. Finally, expanding the scope of research beyond lung cancer to encompass other malignancies and chronic diseases is necessary to determine the broader applicability of T12-based body composition analysis.

Conclusion

This study demonstrates that CT measurements at the T12 level, particularly SMA and SMI, provide a reliable and practical alternative to L3-based assessments for evaluating muscle mass in patients with advanced lung cancer. T12 measurements not only show strong agreement with L3 measurements but also exhibit a higher correlation with DXA-determined trunk LST. While T12 SMD proves to be a suitable substitute for L3 SMD in muscle composition assessment, T12 IMAT does not serve as a reliable alternative. These findings advocate for the integration of T12-level muscle assessment into clinical research as a viable alternative, leveraging the widespread use of thoracic CT scans. Future multicenter prospective studies are warranted to further validate the prognostic significance of T12 measurements and explore their broader applications in various clinical settings.

Declaration of competing interest

The authors declare no conflict of interest.

CRediT authorship contribution statement

Rongna Lian: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Tianjiao Tang:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Wenhua Jiang:** Formal analysis, Data curation. **Jiaojiao Jiang:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Ming Yang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

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Data availability

The raw data used in this article can be obtained from the corresponding author on reasonable request.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.nut.2025.112894.

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